

Mathematical Statistics

Basic Foundation of Probability and Statistics

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1 Introduction & Course Outline

This lecture series aims to build a strong foundation in Probability theory, starting from the absolute basics.

Course Outline

The key topics that will be covered in this series include:

- **Fundamental Terminology:** Random experiments, sample space, events.
- **Types of Events:** Simple, compound, mutually exclusive, exhaustive, independent, etc.
- **Theorems of Probability:** Addition, multiplication, conditional probability, and Bayes' theorem.
- **Random Variables:** Discrete and continuous random variables, PDF, PMF.
- **Expectation and Variance:** Calculating the mean, variance, and standard deviation.
- **Probability Distributions:** In-depth discussion on discrete and continuous distributions.

2 Fundamental Concepts

2.1 Random Experiment

Definition: Random Experiment

A **Random Experiment** is any process or action whose outcome cannot be predicted with certainty, but all possible outcomes are known beforehand. An experiment is considered random if:

1. It can be repeated under identical conditions.
2. The set of all possible outcomes is known.
3. The exact outcome of any specific trial is unknown in advance.

Examples: Tossing a coin, rolling a die, drawing a card from a shuffled deck.

Key Insight

If an outcome is predetermined or planned (e.g., using a leaked exam paper), the concept of probability does not apply. Probability thrives on randomness and unpredictability.

2.2 Sample Space (S)

The Sample Space is the cornerstone of probability. Without knowing the full scope of what *can* happen, we cannot calculate the probability of what *did* or *will* happen.

💡 Definition: Sample Space

The sample space, denoted by S , is the set of **all possible outcomes** of a random experiment. Each individual outcome in the sample space is called a **sample point**.

2.2.1 Constructing Sample Spaces

• Tossing a Coin:

- 1 Toss: $S = \{H, T\}$. Total outcomes = $2^1 = 2$.
- 2 Tosses: $S = \{HH, HT, TH, TT\}$. Total outcomes = $2^2 = 4$.
- 3 Tosses: $S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$. Total outcomes = $2^3 = 8$.

• Rolling a Die:

- 1 Roll: $S = \{1, 2, 3, 4, 5, 6\}$. Total outcomes = $6^1 = 6$.
- 2 Rolls: The sample space consists of 36 ordered pairs. Total outcomes = $6^2 = 36$.

⚙️ Example: Probability with Two Dice

What is the probability that the sum of the numbers on two dice is 11 or more?

- **Sample Space (S):** 36 outcomes, from (1,1) to (6,6).
- **Favorable Outcomes (Event E):** Pairs whose sum is ≥ 11 .
 - Sum = 11: $\{(5, 6), (6, 5)\}$
 - Sum = 12: $\{(6, 6)\}$
- **Total favorable outcomes = 3.**
- **Probability:** $P(E) = \frac{\text{Favorable Outcomes}}{\text{Total Outcomes}} = \frac{3}{36} = \frac{1}{12}$.

2.3 Playing Cards - A Detailed Look

Understanding a standard deck of 52 cards is crucial for many probability problems.

Category	Breakdown	Total Cards
Total Cards	A standard deck	52
Colors	26 Red Cards & 26 Black Cards	52
Suits	Red: 13 Hearts (♥), 13 Diamonds (♦) Black: 13 Spades (♠), 13 Clubs (♣)	4 suits of 13 cards each
Face Cards	Jack (J), Queen (Q), King (K) in each of the 4 suits.	12
Honour Cards	Ace (A), Jack (J), Queen (Q), King (K) in each suit.	16

3 Events and Their Types

💡 Definition: Event (E)

An **Event** is any subset of the sample space ($E \subseteq S$). It represents a specific outcome or a collection of outcomes that we are interested in.

3.1 Types of Events

- **Simple Event:** An event consisting of only **one** sample point. Ex: Tossing two coins, $E = \{HH\}$.
- **Compound Event:** An event with **more than one** sample point. Ex: Tossing two coins, "at least one head" is $E = \{HH, HT, TH\}$.
- **Mutually Exclusive Events:** Two events, A and B, that **cannot occur at the same time**. Mathematically, their intersection is the empty set: $A \cap B = \emptyset$.
- **Exhaustive Events:** A set of events whose union forms the entire sample space. At least one of these events must occur. Mathematically: $A_1 \cup A_2 \cup \dots \cup A_n = S$.
- **Complementary Events:** The complement of an event A, denoted A' or A^c , is the set of all outcomes in S that are not in A. It means "event A does not happen." $A' = S - A$.
- **Independent Events:** The occurrence of one event **does not affect** the probability of the other occurring.
- **Dependent Events:** The occurrence of one event **changes** the probability of the other.

4 Solved Examples

⚙️ Example: Mutually Exclusive and Exhaustive Events

Two dice are rolled. Let A be the event that the sum is 5, and B be the event that at least one die shows a 3. Are A and B mutually exclusive or exhaustive?

Step 1: Define Events A and B. $A = \{(1, 4), (2, 3), (3, 2), (4, 1)\}$

$B = \{(3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), (1, 3), (2, 3), (4, 3), (5, 3), (6, 3)\}$

Step 2: Check for Mutually Exclusive property. Find the intersection $A \cap B$.

$A \cap B = \{(2, 3), (3, 2)\}$. Since $A \cap B \neq \emptyset$, the events are **NOT mutually exclusive**.

Step 3: Check for Exhaustive property. Find the union $A \cup B$.

$|A \cup B| = |A| + |B| - |A \cap B| = 4 + 11 - 2 = 13$. Since the union contains only 13 of the 36 possible outcomes, it does not equal the sample space S. The events are **NOT exhaustive**.

⚙️ Example: Constructing a Conditional Sample Space

A coin is tossed. If it shows heads (H), a ball is drawn from a bag with 3 blue (B) and 4 white (W) balls. If it shows tails (T), a die is rolled. Describe the sample space.

Heads Branch: If the coin is H, we draw a ball. Outcomes:
 $\{(H, B_1), (H, B_2), (H, B_3), (H, W_1), (H, W_2), (H, W_3), (H, W_4)\}$.

Tails Branch: If the coin is T, we roll a die. Outcomes:
 $\{(T, 1), (T, 2), (T, 3), (T, 4), (T, 5), (T, 6)\}$.

Full Sample Space (S): The union of outcomes from both branches.

$S = \{(H, B_1), \dots, (H, W_4), (T, 1), \dots, (T, 6)\}$.

5 Summary & Practice

Summary & Key Takeaways

- **Random Experiment:** A process with an uncertain outcome.
- **Sample Space (S):** The set of all possible outcomes.
- **Event (E):** A subset of the sample space ($E \subseteq S$).
- **Mutually Exclusive:** Events that cannot happen together ($A \cap B = \emptyset$).
- **Exhaustive:** Events whose union is the entire sample space ($A \cup B = S$).

Practice Problems

1. A single card is drawn from a standard 52-card deck. Let E be the event of drawing a face card and F be the event of drawing a spade.
 - (a) List the sample points in events E and F.
 - (b) Are events E and F mutually exclusive? Explain why or why not.
2. An experiment consists of rolling a die. If the number is even, a coin is flipped once. If the number is odd, the coin is flipped twice. Write out the entire sample space for this experiment.
3. A coin is tossed. If it shows a head, we draw a ball from a bag with 3 blue and 4 white balls. If it shows a tail, we throw a die. Let A be the event "a blue ball is drawn" and B be the event "an even number appears on the die". Are A and B mutually exclusive? Explain your reasoning.

6 Introduction

This lecture2, we will explore the core of probability calculation, the concepts of "odds," and one of its most important rules: the Addition Theorem.

Course Outline

Today's Key Topics:

- The classical definition and calculation of probability.
- Fundamental properties and rules governing probability.
- The concepts of "Odds in Favor" and "Odds Against."
- The Addition Theorem for both mutually exclusive and non-mutually exclusive events.

7 The Classical Definition of Probability

Definition: Definition: Probability

If a random experiment has a finite sample space S with n equally likely outcomes, and an event E is a subset of S with m favorable outcomes, then the probability of event E is:

$$P(E) = \frac{\text{Number of Favorable Outcomes for } E}{\text{Total Number of Possible Outcomes}} = \frac{n(E)}{n(S)} = \frac{m}{n}$$

Example: Drawing Balls from a Bag

A bag contains 7 white and 5 black balls.

- What is the probability of drawing one white ball?
- What is the probability of drawing two balls: one white and one black?

Solution:

Part a) Total balls = 12. Total ways to draw 1 ball = ${}^{12}C_1 = 12$. Ways to draw 1 white ball = ${}^7C_1 = 7$.

$$P(1 \text{ White}) = \frac{{}^7C_1}{{}^{12}C_1} = \frac{7}{12}$$

Part b) Total ways to draw 2 balls from 12 = ${}^{12}C_2 = 66$. Ways to draw 1 white AND 1 black ball = ${}^7C_1 \times {}^5C_1 = 35$.

$$P(1W \text{ and } 1B) = \frac{{}^7C_1 \times {}^5C_1}{{}^{12}C_2} = \frac{35}{66}$$

8 Fundamental Properties of Probability

- **Probability Range:** For any event E , $0 \leq P(E) \leq 1$.
- **Certain Event:** The probability of the entire sample space is 1, i.e., $P(S) = 1$.
- **Impossible Event:** The probability of an impossible event (empty set) is 0, i.e., $P(\emptyset) = 0$.

Key Insight

The Complement Rule If $P(A)$ is the probability that event A occurs, then the probability it does not occur is $P(A')$.

$$P(A) + P(A') = 1 \implies P(A') = 1 - P(A)$$

This rule is extremely useful for calculating the probability of "at least one" or "not all" events.

9 Odds in Favor and Odds Against

💡 Definition: Odds

Odds express likelihood as a ratio of favorable to unfavorable outcomes.

- **Odds in Favor of E** = $\frac{\text{Favorable Outcomes}}{\text{Unfavorable Outcomes}} = \frac{n(E)}{n(S)-n(E)}$
- **Odds Against E** = $\frac{\text{Unfavorable Outcomes}}{\text{Favorable Outcomes}} = \frac{n(S)-n(E)}{n(E)}$

⚙️ Example: Odds in a Die Roll

When a die is rolled, what are the odds in favor of getting a number greater than 4?

Sample Space: $S = \{1, 2, 3, 4, 5, 6\}$.

Event E: "Number > 4" = $\{5, 6\}$. Favorable outcomes = 2.

Unfavorable Outcomes: $\{1, 2, 3, 4\}$. Unfavorable outcomes = 4.

Calculation: Odds in Favor = $\frac{2}{4} = \frac{1}{2}$ (or 1:2).

10 The Addition Theorem of Probability

This theorem helps find the probability of at least one of two events occurring (i.e., $P(A \text{ or } B)$).

10.1 For Mutually Exclusive Events

If events A and B cannot occur simultaneously ($A \cap B = \emptyset$), the formula is:

$$P(A \cup B) = P(A) + P(B)$$

⚙️ Example: Lottery Tickets (Mutually Exclusive)

A lottery has tickets numbered 1 to 20. Find the probability that a drawn ticket is a multiple of 5 OR a multiple of 7.

Events: A = "Multiple of 5" = $\{5, 10, 15, 20\}$, so $P(A) = 4/20$.

B = "Multiple of 7" = $\{7, 14\}$, so $P(B) = 2/20$.

Intersection: $A \cap B = \emptyset$. The events are mutually exclusive.

Probability: $P(A \cup B) = P(A) + P(B) = \frac{4}{20} + \frac{2}{20} = \frac{6}{20} = \frac{3}{10}$.

10.2 For Non-Mutually Exclusive Events (General Rule)

If events A and B can occur simultaneously ($A \cap B \neq \emptyset$), we subtract the intersection to avoid double-counting.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

⚙️ Example: Rolling Two Dice

Find the probability of getting either a doublet OR a total of 10.

Events: $A = \text{"Doublet"}, P(A) = 6/36$. $B = \text{"Total of 10"}, P(B) = 3/36$.

Intersection: $A \cap B = \{(5, 5)\}$, so $P(A \cap B) = 1/36$.

Probability: $P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{6}{36} + \frac{3}{36} - \frac{1}{36} = \frac{8}{36} = \frac{2}{9}$.

11 Summary Practice

📖 Summary & Key Takeaways

- **Probability:** A measure of likelihood, calculated as Favorable / Total.
- **Complement Rule:** $P(A') = 1 - P(A)$ is essential for "at least one" or "not all" problems.
- **Addition Theorem (OR rule):** Use $P(A) + P(B)$ for mutually exclusive events and $P(A) + P(B) - P(A \cap B)$ for non-mutually exclusive events.

1. A number is chosen at random from the first 50 positive integers. What is the probability that the number is a multiple of 4 or a multiple of 5?
2. Two cards are drawn from a standard deck of 52 cards. What is the probability that both cards are face cards (Jack, Queen, King)?
3. The probability of a student passing an exam is 0.6. What are the odds against the student passing the exam?
4. A hand of 7 cards is dealt from a shuffled 52-card deck. Find the probability the hand contains:
 1. Exactly three Kings.
 2. At least one King. (*Hint: Use the complement rule.*)

12 Lecture Overview

This lecture introduces the concept of **independent events** in probability. We will explore the fundamental **Multiplication Theorem** for calculating the probability of two or more independent events occurring. A key takeaway will be a powerful deduction for solving "at least one" type problems, which is a common and important application. This topic serves as a foundation for understanding more complex concepts like dependent events and conditional probability, which will be covered in subsequent lectures.

Understanding independent events is crucial for modeling real-world scenarios where outcomes don't influence each other. From analyzing system reliability (will a machine with multiple parts work?) to assessing risk in financial markets or even determining the outcome of games, the principles learned here are universally applicable. Mastering the multiplication rule and the "at least one" shortcut provides a powerful toolkit for any student in mathematics, engineering, or data science.

13 Key Concepts

Definition:

13.1

Independent Events Two or more events are considered **independent** if the occurrence or non-occurrence of one event has no effect on the probability of the other events occurring.

- **Example:** Flipping a coin twice. The outcome of the first flip (Heads or Tails) does not influence the outcome of the second flip.

13.2 Multiplication Theorem for Independent Events

If events A and B are independent, the probability that **both** events occur is the product of their individual probabilities. This is also known as the probability of the intersection of A and B .

$$P(A \text{ and } B) = P(A \cap B) = P(A) \times P(B)$$

This can be extended to three or more independent events:

$$P(A \cap B \cap C) = P(A) \times P(B) \times P(C)$$

13.3 Complementary Events

The complement of an event A , denoted as A' (or A^c or $\bar{A}$), is the event that A does not occur. The sum of the probabilities of an event and its complement is always 1.

$$P(A') = 1 - P(A)$$

This is crucial for calculating the probability of an event *not* happening.

13.4 Important Deduction: The "At Least One" Rule

A much simpler method for finding the probability that *at least one* event occurs is to use the complement rule. The opposite of "at least one event occurs" is **"none of the events occur."**

$$P(\text{At least one event occurs}) = 1 - P(\text{None of the events occur})$$

For three independent events X , Y , and Z :

$$\begin{aligned} P(\text{At least one of } X, Y, Z) &= 1 - P(X' \cap Y' \cap Z') \\ &= 1 - [P(X') \times P(Y') \times P(Z')] \end{aligned}$$

14 Solved Examples

Here are the step-by-step solutions to the problems discussed in the lecture.

⚙️ Example: Example 1: The Statistics Problem

Problem: A statistics problem is given to three students: A, B, and C. Their respective probabilities of solving it are $\frac{1}{2}$, $\frac{1}{3}$, and $\frac{1}{4}$. What is the probability that the problem will be solved?

Solution:

Step 1: Identify the Goal: "The problem will be solved" means that **at least one** of the students solves it.

Step 2: Define Events:

$$P(A) = \frac{1}{2}, \quad P(B) = \frac{1}{3}, \quad P(C) = \frac{1}{4}$$

Step 3: Apply the "At Least One" Rule:

$$P(\text{Problem solved}) = 1 - P(\text{Problem NOT solved})$$

The problem is not solved only if A, B, AND C all fail to solve it.

$$P(\text{Problem solved}) = 1 - P(A' \cap B' \cap C')$$

Step 4: Calculate Probabilities of Failure (Complements):

$$P(A') = 1 - P(A) = 1 - \frac{1}{2} = \frac{1}{2}$$

$$P(B') = 1 - P(B) = 1 - \frac{1}{3} = \frac{2}{3}$$

$$P(C') = 1 - P(C) = 1 - \frac{1}{4} = \frac{3}{4}$$

Step 5: Use the Multiplication Theorem:

$$P(A' \cap B' \cap C') = P(A') \times P(B') \times P(C') = \frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} = \frac{6}{24} = \frac{1}{4}$$

Step 6: Final Calculation:

$$P(\text{Problem solved}) = 1 - \frac{1}{4} = \frac{3}{4}$$

Answer: The probability that the problem will be solved is $\frac{3}{4}$.

⚙️ Example: Example 2: The Contradiction Problem

Problem: A speaks the truth in 75% of cases, and B speaks the truth in 80% of cases. What is the probability that they will contradict each other when stating the same fact?

Solution:

Step 1: Identify the Goal: Contradiction occurs when one person tells the truth and the other lies.

Step 2: Define Events & Probabilities:

$$P(A) = P(\text{A speaks truth}) = 75\% = \frac{75}{100} = \frac{3}{4}$$
$$P(B) = P(\text{B speaks truth}) = 80\% = \frac{80}{100} = \frac{4}{5}$$

Step 3: Calculate Complementary Probabilities (Lying):

$$P(A') = P(\text{A lies}) = 1 - P(A) = 1 - \frac{3}{4} = \frac{1}{4}$$
$$P(B') = P(\text{B lies}) = 1 - P(B) = 1 - \frac{4}{5} = \frac{1}{5}$$

Step 4: Identify the Cases for Contradiction: There are two mutually exclusive possibilities:

- **Case 1:** A speaks the truth AND B lies ($A \cap B'$).
- **Case 2:** A lies AND B speaks the truth ($A' \cap B$).

Step 5: Calculate Probability of Each Case:

$$P(\text{Case 1}) = P(A \cap B') = P(A) \times P(B') = \frac{3}{4} \times \frac{1}{5} = \frac{3}{20}$$
$$P(\text{Case 2}) = P(A' \cap B) = P(A') \times P(B) = \frac{1}{4} \times \frac{4}{5} = \frac{4}{20}$$

Step 6: Final Calculation: Sum the probabilities of the cases.

$$P(\text{Contradiction}) = P(\text{Case 1}) + P(\text{Case 2}) = \frac{3}{20} + \frac{4}{20} = \frac{7}{20}$$

Answer: The probability that they will contradict each other is $\frac{7}{20}$.

⚙️ Example: Example 3: The Target Hitting Problem

Problem: A hits a target 4 times in 5 shots, B hits 3 times in 4 shots, and C hits 2 times in 3 shots. Find the probability that:

- a) The target is hit.
- b) The target is hit at least twice.

Solution (Part a): The target is hit

Step 1: Define Probabilities of Hitting:

$$P(A) = \frac{4}{5}, \quad P(B) = \frac{3}{4}, \quad P(C) = \frac{2}{3}$$

Step 2: Calculate Probabilities of Missing:

$$P(A') = 1 - \frac{4}{5} = \frac{1}{5}, \quad P(B') = 1 - \frac{3}{4} = \frac{1}{4}, \quad P(C') = 1 - \frac{2}{3} = \frac{1}{3}$$

Step 3: Apply "At Least One" Rule:

$$\begin{aligned} P(\text{Target hit}) &= 1 - P(\text{No one hits}) \\ &= 1 - P(A' \cap B' \cap C') \\ &= 1 - [P(A') \times P(B') \times P(C')] \\ &= 1 - \left(\frac{1}{5} \times \frac{1}{4} \times \frac{1}{3} \right) = 1 - \frac{1}{60} = \frac{59}{60} \end{aligned}$$

Answer (a): The probability that the target is hit is $\frac{59}{60}$.

Solution (Part b): The target is hit at least twice

Step 1: Identify Favorable Cases: "At least twice" means either exactly two hits OR all three hit.

- Case 1: A and B hit, C misses ($A \cap B \cap C'$)
- Case 2: A and C hit, B misses ($A \cap B' \cap C$)
- Case 3: B and C hit, A misses ($A' \cap B \cap C$)
- Case 4: A, B, and C all hit ($A \cap B \cap C$)

Step 2: Calculate Probability of Each Case:

$$\begin{aligned} P(\text{Case 1}) &= P(A) \times P(B) \times P(C') = \frac{4}{5} \times \frac{3}{4} \times \frac{1}{3} = \frac{12}{60} \\ P(\text{Case 2}) &= P(A) \times P(B') \times P(C) = \frac{4}{5} \times \frac{1}{4} \times \frac{2}{3} = \frac{8}{60} \\ P(\text{Case 3}) &= P(A') \times P(B) \times P(C) = \frac{1}{5} \times \frac{3}{4} \times \frac{2}{3} = \frac{6}{60} \\ P(\text{Case 4}) &= P(A) \times P(B) \times P(C) = \frac{4}{5} \times \frac{3}{4} \times \frac{2}{3} = \frac{24}{60} \end{aligned}$$

Step 3: Final Calculation: Sum the probabilities.

$$P(\text{At least 2 hits}) = \frac{12}{60} + \frac{8}{60} + \frac{6}{60} + \frac{24}{60} = \frac{50}{60} = \frac{5}{6}$$

Answer (b): The probability of at least two hits is $\frac{5}{6}$.

⚙️ Example: Example 4: The Groups of Children Problem

Problem: There are three groups:

- Group 1: 3 Girls, 1 Boy (Total: 4)
- Group 2: 2 Girls, 2 Boys (Total: 4)
- Group 3: 1 Girl, 3 Boys (Total: 4)

One child is selected from each group. What is the probability of selecting exactly 1 girl and 2 boys?

Solution:

Step 1: Identify Favorable Cases: We need 1 Girl and 2 Boys. This can happen in three ways:

- **Case 1:** Girl from G1, Boy from G2, Boy from G3 (G_1, B_2, B_3)
- **Case 2:** Boy from G1, Girl from G2, Boy from G3 (B_1, G_2, B_3)
- **Case 3:** Boy from G1, Boy from G2, Girl from G3 (B_1, B_2, G_3)

Step 2: Calculate Probability of Each Case:

$$P(\text{Case 1}) = P(G_1) \times P(B_2) \times P(B_3) = \frac{3}{4} \times \frac{2}{4} \times \frac{3}{4} = \frac{18}{64}$$

$$P(\text{Case 2}) = P(B_1) \times P(G_2) \times P(B_3) = \frac{1}{4} \times \frac{2}{4} \times \frac{3}{4} = \frac{6}{64}$$

$$P(\text{Case 3}) = P(B_1) \times P(B_2) \times P(G_3) = \frac{1}{4} \times \frac{2}{4} \times \frac{1}{4} = \frac{2}{64}$$

Step 3: Final Calculation: Sum the probabilities of these mutually exclusive cases.

$$P(1 \text{ Girl, } 2 \text{ Boys}) = \frac{18}{64} + \frac{6}{64} + \frac{2}{64} = \frac{26}{64} = \frac{13}{32}$$

Answer: The probability of selecting 1 girl and 2 boys is $\frac{13}{32}$.

15 Summary and Exercises

Summary & Key Takeaways

- **Independent Events:** The outcome of one event does not affect the outcome of another.
- **Multiplication Rule (AND):** For independent events, $P(A \cap B) = P(A) \times P(B)$. Use this when you need multiple specific outcomes to happen simultaneously.
- **Addition Rule (OR):** For mutually exclusive events, $P(A \cup B) = P(A) + P(B)$. Use this when you have different cases that all satisfy the desired outcome.
- **The "At Least One" Shortcut:** This is one of the most powerful tools for these problems. It's almost always easier to calculate $1 - P(\text{None})$ than to calculate the probability of "at least one" directly.

Test your understanding with these problems.

1. **(From the Lecture):** A machine has three components: A, B, and C. The probabilities of them failing within a year are $\frac{1}{5}$, $\frac{1}{3}$, and $\frac{1}{2}$, respectively. What is the probability that the machine will be working at the end of the year? (Hint: The machine works only if NONE of its components fail).
2. Two fair six-sided dice are rolled. What is the probability that the first die shows a 6 AND the second die shows a 3?
3. A bag contains 5 red and 3 blue balls. Another bag contains 4 red and 6 blue balls. If one ball is drawn from each bag, what is the probability that:
 - a) Both balls are red?
 - b) At least one ball is blue?
4. The probability of a husband and wife being alive in 20 years is 0.8 and 0.9, respectively. Find the probability that in 20 years:
 - a) Both will be alive.
 - b) Neither will be alive.
 - c) At least one of them will be alive.

16 Lecture Overview

This lecture transitions from independent events to the more nuanced concept of **dependent events**. When the outcome of one event influences the probability of another, we enter the realm of **conditional probability**. We will define conditional probability, introduce its notation and core formula, and see how it leads to a more general form of the Multiplication Theorem. This concept is fundamental for solving problems where events are sequential and occur "without replacement," and it lays the essential groundwork for understanding advanced topics like Bayes' Theorem.

Conditional probability is the mathematical engine behind reasoning with new information. It's used everywhere, from medical diagnoses (the probability of a disease given certain symptoms) to machine learning (updating a model's beliefs based on new data) and simple games (your next move in a card game depends on the cards already

played). Mastering this concept allows you to build models that adapt and respond to an ever-changing world.

17 Key Concepts and Formulas

💡 Definition:

17.1

Dependent Events Two events are **dependent** if the occurrence of one event changes the probability of the other event occurring.

- **Example:** Drawing two cards from a deck *without replacement*. After drawing the first card, the composition of the remaining deck changes, altering the probabilities for the second draw.

17.2 Conditional Probability

Conditional probability is the probability of an event (say, A) occurring, given that another event (B) has already occurred.

Notation: $P(A|B)$

Read as: "The probability of A , given B ."

It answers: "Knowing that event B has happened, what is now the likelihood of event A happening?"

The probability of event A occurring given that event B has occurred is calculated as:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Similarly, the probability of B given A is: $P(B|A) = \frac{P(A \cap B)}{P(A)}$. **Note:** The denominator is always the probability of the event that is *known* to have occurred.

17.3 The General Multiplication Rule (for Dependent Events)

By rearranging the conditional probability formula, we derive the general multiplication rule, which works for both dependent and independent events.

$$P(A \cap B) = P(A) \times P(B|A)$$

This reads: The probability of A and B both happening is the probability of A happening, multiplied by the probability of B happening *given that A has already happened*. Alternatively: $P(A \cap B) = P(B) \times P(A|B)$.

17.4 Relationship to Independent Events

For **independent events**, the occurrence of A does not affect B , so $P(B|A) = P(B)$. Substituting this into the general rule gives $P(A \cap B) = P(A) \times P(B)$, which is the rule we learned previously. This shows that the rule for independent events is a special case of this more general rule.

18 Solved Examples

🔧 Example: Example 1: The Dice Problem

Problem: If a fair die is tossed and an odd number comes up, what is the probability of it being a prime number?

Solution:

Step 1: Define Sample Space and Events:

- Total Sample Space, $S = \{1, 2, 3, 4, 5, 6\}$
- Let A be the event that an odd number comes up: $A = \{1, 3, 5\}$
- Let B be the event that a prime number comes up: $B = \{2, 3, 5\}$

Step 2: Identify the Goal: We are given that event A has occurred. We need to find the probability of event B . We need to find $P(B|A)$.

Step 3: Apply the Formula: $P(B|A) = \frac{P(A \cap B)}{P(A)}$

Step 4: Calculate Necessary Probabilities:

- $P(A)$: Probability of getting an odd number is $P(A) = \frac{3}{6}$.
- $A \cap B$: The numbers that are both odd and prime are $\{3, 5\}$.
- $P(A \cap B)$: Probability of being both odd and prime is $P(A \cap B) = \frac{2}{6}$.

Step 5: Final Calculation: $P(B|A) = \frac{2/6}{3/6} = \frac{2}{3}$

Intuitive Approach: Once we know an odd number has occurred, our sample space shrinks to $\{1, 3, 5\}$. Within this new space, the prime numbers are $\{3, 5\}$. This gives 2 favorable outcomes out of 3 total possibilities. Hence, the probability is $\frac{2}{3}$.

⚙️ Example: Example 2: Failing Students

Problem: In a school, 20% of students failed in English, 15% failed in Mathematics, and 10% failed in both. A student is selected at random. If they failed in Mathematics, what is the probability they also failed in English?

Solution:

Step 1: Define Events and Given Probabilities:

- Let E be the event a student failed in English: $P(E) = 0.20$
- Let M be the event a student failed in Mathematics: $P(M) = 0.15$
- The event a student failed in both: $P(E \cap M) = 0.10$

Step 2: Identify the Goal: We are given M has occurred. We need to find the probability of E . We are finding $P(E|M)$.

Step 3: Apply the Formula: $P(E|M) = \frac{P(E \cap M)}{P(M)}$

Step 4: Final Calculation: $P(E|M) = \frac{0.10}{0.15} = \frac{10}{15} = \frac{2}{3}$

Answer: The probability is $\frac{2}{3}$.

⚙️ Example: Example 3: Balls in an Urn (Without Replacement)

Problem: An urn contains 10 white and 5 black balls. Two balls are drawn one after the other without replacement. What is the probability that both balls drawn are white?

Solution:

Step 1: Define Events:

- Let W_1 be the event that the first ball is white.
- Let W_2 be the event that the second ball is white.

Step 2: Identify the Goal: Find the probability that the first is white AND the second is white: $P(W_1 \cap W_2)$.

Step 3: Apply the General Multiplication Rule: $P(W_1 \cap W_2) = P(W_1) \times P(W_2|W_1)$.

Step 4: Calculate Each Probability:

- $P(W_1) = \frac{\text{White Balls}}{\text{Total Balls}} = \frac{10}{15}$.
- $P(W_2|W_1)$: After one white ball is drawn, 9 white and 14 total balls remain. Thus, $P(W_2|W_1) = \frac{9}{14}$.

Step 5: Final Calculation: $P(W_1 \cap W_2) = \frac{10}{15} \times \frac{9}{14} = \frac{2}{3} \times \frac{9}{14} = \frac{18}{42} = \frac{3}{7}$.

Answer: The probability of drawing two white balls is $\frac{3}{7}$.

⚙️ Example: Example 4: Drawing Cards (Without Replacement)

Problem: Two cards are drawn one by one without replacement from a pack of 52 cards. What is the probability that one is a Red Queen and the other is a Black King?

Solution:

Step 1: Analyze the Problem: The order is not specified, creating two mutually exclusive cases:

- **Case 1:** Draw a Red Queen first (RQ), then a Black King (BK).
- **Case 2:** Draw a Black King first (BK), then a Red Queen (RQ).

The total probability is $P(\text{Case 1}) + P(\text{Case 2})$. There are 2 Red Queens and 2 Black Kings in a deck.

Step 2: Calculate Probability of Case 1 (RQ then BK):

$$P(\text{Case 1}) = P(RQ_1) \times P(BK_2|RQ_1) = \frac{2}{52} \times \frac{2}{51} = \frac{4}{2652}$$

Step 3: Calculate Probability of Case 2 (BK then RQ):

$$P(\text{Case 2}) = P(BK_1) \times P(RQ_2|BK_1) = \frac{2}{52} \times \frac{2}{51} = \frac{4}{2652}$$

Step 4: Final Calculation: Sum the probabilities.

$$P(\text{Total}) = P(\text{Case 1}) + P(\text{Case 2}) = \frac{4}{2652} + \frac{4}{2652} = \frac{8}{2652} = \frac{2}{663}$$

Answer: The probability is $\frac{2}{663}$.

19 Summary and Exercises

Summary & Key Takeaways

- **Conditional Probability** is used for **dependent events**, where one event's outcome affects another's probability.
- The core formula is $P(A|B) = \frac{P(A \cap B)}{P(B)}$, calculating the probability of A given B has occurred.
- **"Without Replacement"** is a key phrase indicating that events are dependent.
- The **General Multiplication Rule**, $P(A \cap B) = P(A) \times P(B|A)$, is essential for finding the probability of a sequence of dependent events.
- When a problem does not specify the order of events, remember to calculate the probability for **all possible orderings** and add them together.

1. A pair of fair dice is rolled. If the sum is 6, what is the probability that one of the dice shows a 2?
2. In a class, 40% of the students study math and science. 60% of the students study math. What is the probability of a student studying science given they study math?
3. A bag contains 5 red and 7 green marbles. Two marbles are drawn one after another without replacement. Find the probability that:
 - a) The first is red and the second is green.
 - b) Both marbles are red.
4. From a standard deck of 52 cards, two cards are drawn without replacement. What is the probability of drawing two Aces?
5. A car factory has two assembly lines. Line A produces 60% of the cars and Line B produces 40%. 5% of cars from Line A have defects, and 8% from Line B have defects. If a car selected at random is found to have a defect, what is the probability it came from Line A? (Advanced problem leading towards Bayes' Theorem).

20 Lecture Overview

This lecture delves into one of the most important concepts in probability theory: **Bayes' Theorem**. Building upon the foundation of conditional probability, we will explore how to calculate the probability of an event by considering all possible scenarios that could lead to it (the Law of Total Probability). We will then introduce Bayes' Theorem itself, a powerful tool for calculating **inverse probability**—that is, determining the probability of an initial cause given that we have observed a particular final outcome. This allows us to update our beliefs in light of new evidence.

Bayes' Theorem formalizes how we learn from experience. It's the mathematical rule for updating our beliefs when we get new data. This idea is central to fields like artificial intelligence (spam filters use it to decide if an email is junk), medical diagnosis (updating the chance of a disease after a test result), and scientific research (assessing how a new experiment supports a hypothesis). Understanding it means you can reason rigorously about uncertainty and make smarter, data-driven decisions.

21 Theoretical Foundations

💡 Definition:

21.1

Partition of a Sample Space Imagine a sample space S that is completely divided into several **mutually exclusive** and **exhaustive** events, $B_1, B_2, \dots, B_n$.

- **Mutually Exclusive:** No two events can happen at the same time ($B_i \cap B_j = \emptyset$ for $i \neq j$).
- **Exhaustive:** The union of all these events covers the entire sample space ($B_1 \cup B_2 \cup \dots \cup B_n = S$).

These events form a **partition** of the sample space, representing all possible initial scenarios or causes.

21.2 The Law of Total Probability

The Law of Total Probability allows us to find the probability of an event A (an "effect") by summing the probabilities of A occurring through each possible cause (B_i).

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \dots + P(A|B_n)P(B_n)$$

In a compact form:

$$P(A) = \sum_{i=1}^n P(A|B_i)P(B_i)$$

This formula gives the overall probability of an effect before we know the specific cause.

21.3 Bayes' Theorem (Inverse Probability)

Bayes' Theorem tells us how to calculate the probability of a specific cause (B_i) given that we have already observed an effect (A). It "inverts" the conditional probability from $P(A|B)$ to $P(B|A)$.

$$P(B_i|A) = \frac{P(A \cap B_i)}{P(A)} = \frac{P(A|B_i)P(B_i)}{P(A)}$$

Substituting the Law of Total Probability for $P(A)$ gives the full form:

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{\sum_{j=1}^n P(A|B_j)P(B_j)}$$

In words: $P(\text{Cause}|\text{Effect}) = \frac{P(\text{Effect}|\text{Cause}) \times P(\text{Cause})}{P(\text{Total Probability of Effect})}$

22 Solved Examples

Example: Example 1: The Defective Bolts Problem

Problem: A company buys bolts from three vendors: A (30%), B (40%), and C (30%). Known defect rates are: A (2%), B (3%), C (5%).

- What is the probability that a randomly selected bolt is defective?
- If a bolt is defective, what is the probability it came from vendor A?

Solution:

Step 1: Define Events & Probabilities:

- A, B, C : Bolt is from Vendor A, B, C. D : Bolt is defective.
- Priors:** $P(A) = 0.30$, $P(B) = 0.40$, $P(C) = 0.30$.
- Likelihoods:** $P(D|A) = 0.02$, $P(D|B) = 0.03$, $P(D|C) = 0.05$.

Step 2: Part (a) - Total Probability of a Defect $P(D)$:

$$\begin{aligned}P(D) &= P(D|A)P(A) + P(D|B)P(B) + P(D|C)P(C) \\&= (0.02)(0.30) + (0.03)(0.40) + (0.05)(0.30) \\&= 0.006 + 0.012 + 0.015 = \mathbf{0.033}\end{aligned}$$

Step 3: Part (b) - Inverse Probability $P(A|D)$: Use Bayes' Theorem.

$$P(A|D) = \frac{P(D|A)P(A)}{P(D)} = \frac{0.006}{0.033} = \frac{6}{33} = \frac{\mathbf{2}}{\mathbf{11}}$$

Example: Example 2: The Professor Promotion Problem

Problem: The chances of candidates X, Y, Z getting a promotion are in the ratio 4:2:3. The probabilities they will introduce a new subject if selected are 0.35, 0.52, and 0.80, respectively. What is the probability a new subject will be introduced?

Solution:

Step 1: Define Events & Probabilities:

- X, Y, Z : Candidate X, Y, or Z is selected. N : A new subject is introduced.
- Priors:** Total ratio parts = $4 + 2 + 3 = 9$. $P(X) = 4/9$, $P(Y) = 2/9$, $P(Z) = 3/9$.
- Likelihoods:** $P(N|X) = 0.35$, $P(N|Y) = 0.52$, $P(N|Z) = 0.80$.

Step 2: Calculate Total Probability $P(N)$:

$$\begin{aligned}P(N) &= P(N|X)P(X) + P(N|Y)P(Y) + P(N|Z)P(Z) \\&= (0.35) \left(\frac{4}{9}\right) + (0.52) \left(\frac{2}{9}\right) + (0.80) \left(\frac{3}{9}\right) \\&= \frac{1}{9}[(0.35 \times 4) + (0.52 \times 2) + (0.80 \times 3)] \\&= \frac{1}{9}[1.40 + 1.04 + 2.40] = \frac{4.84}{9} = \frac{484}{900} = \frac{\mathbf{121}}{\mathbf{225}}\end{aligned}$$

⚙️ Example: Example 3: The Three Boxes Problem

Problem: Three boxes contain balls as follows: B1(1W,2R,3B), B2(2W,3R,1B), B3(3W,1R,2B). A box is chosen at random and two balls are drawn, which are 1 White and 1 Red. What is the probability they came from Box 1?

Solution:

Step 1: Define Events & Probabilities:

- B_1, B_2, B_3 : Box 1, 2, or 3 is chosen. E : 1 White and 1 Red ball are drawn.
- **Priors:** $P(B_1) = P(B_2) = P(B_3) = 1/3$.
- **Likelihoods:** (Total ways to draw 2 from 6 is ${}^6C_2 = 15$) $P(E|B_1) = \frac{{}^1C_1 \times {}^2C_1}{{}^6C_2} = \frac{2}{15}$. $P(E|B_2) = \frac{{}^2C_1 \times {}^3C_1}{{}^6C_2} = \frac{6}{15}$. $P(E|B_3) = \frac{{}^3C_1 \times {}^1C_1}{{}^6C_2} = \frac{3}{15}$.

Step 2: Apply Bayes' Theorem to find $P(B_1|E)$:

$$P(B_1|E) = \frac{P(E|B_1)P(B_1)}{P(E|B_1)P(B_1) + P(E|B_2)P(B_2) + P(E|B_3)P(B_3)}$$

$$= \frac{(2/15)(1/3)}{(2/15)(1/3) + (6/15)(1/3) + (3/15)(1/3)} = \frac{2/45}{11/45} = \frac{2}{11}$$

⚙️ Example: Example 4: The Collapsing Bridge Problem

Problem: A bridge collapse is due to either Wrong Design (WD) or Inferior Material (IM). The chance of wrong design is 10%. If the design is faulty, the probability of collapse is 95%. If the material is inferior, the probability of collapse is 45%. Given the bridge collapsed, what is the chance it was due to wrong designing?

Solution:

Step 1: Define Events & Probabilities:

- WD : Wrong Design, IM : Inferior Material, C : Collapse.
- **Priors:** $P(WD) = 0.10$. Since these are the only two causes, $P(IM) = 1 - 0.10 = 0.90$.
- **Likelihoods:** $P(C|WD) = 0.95$, $P(C|IM) = 0.45$.

Step 2: Goal: Find $P(WD|C)$ using Bayes' Theorem.

$$P(WD|C) = \frac{P(C|WD)P(WD)}{P(C|WD)P(WD) + P(C|IM)P(IM)}$$

$$= \frac{(0.95)(0.10)}{(0.95)(0.10) + (0.45)(0.90)} = \frac{0.095}{0.095 + 0.405} = \frac{0.095}{0.500} = 0.19$$

⚙️ Example: Example 5: The True and False Coins Problem

Problem: There are 3 true coins (H/T) and 1 false coin (H/H). A coin is chosen randomly and tossed 4 times. All 4 tosses result in heads. What is the probability that the false coin was chosen? **Solution:**

Step 1: Define Events & Probabilities:

- T : A true coin is chosen. F : The false coin is chosen. $HHHH$: Four heads are observed.
- **Priors:** $P(T) = 3/4$, $P(F) = 1/4$.
- **Likelihoods:** $P(HHHH|T) = (1/2)^4 = 1/16$. $P(HHHH|F) = (1)^4 = 1$.

Step 2: Goal: Find $P(F|HHHH)$ using Bayes' Theorem.

$$\begin{aligned} P(F|HHHH) &= \frac{P(HHHH|F)P(F)}{P(HHHH|F)P(F) + P(HHHH|T)P(T)} \\ &= \frac{(1)(1/4)}{(1)(1/4) + (1/16)(3/4)} = \frac{1/4}{1/4 + 3/64} = \frac{1/4}{(16 + 3)/64} = \frac{1/4}{19/64} = \frac{1}{4} \times \frac{64}{19} = \frac{16}{19} \end{aligned}$$

23 Summary and Exercises

Summary & Key Takeaways

- **Law of Total Probability:** Used to find the overall probability of an effect by summing its probabilities across all possible causes. $P(A) = \sum P(A|B_i)P(B_i)$.
- **Bayes' Theorem:** Used to find the probability of a specific cause (B_i) given that an effect (A) has already been observed. It "inverts" the conditional probability.
- **Key Application:** Bayes' Theorem is the cornerstone of statistical inference. It allows us to update a prior belief ($P(B_i)$) into a posterior belief ($P(B_i|A)$) after collecting new evidence (A).

1. **(From the Lecture):** I request that you solve it and comment with your answer. Three true coins and one false coin (head on both sides). A coin is chosen and tossed four times. If HEAD occurs all four times, what is the probability that it was a TRUE coin that was chosen? (Hint: Use the results from Example 5).
2. An insurance company insured 2000 scooter drivers, 4000 car drivers, and 6000 truck drivers. The probability of an accident is 0.01, 0.03, and 0.15 respectively. One of the insured persons meets with an accident. What is the probability that he is a scooter driver?
3. A test for a rare disease is 99% accurate. The disease affects 1 in 10,000 people. If a person tests positive, what is the actual probability they have the disease?
4. Two urns contain balls. Urn I: 3 red, 5 black. Urn II: 2 red, 7 black. An urn is chosen at random and a ball is drawn. If the ball is red, what is the probability it came from Urn I?